RAI MUHAMMAD UMAIS

Full-Stack & AI Developer
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PROFESSIONAL SUMMARY

Software Engineering student at FAST-NUCES Lahore (graduating July 2028) who designs, builds, deploys and supports production software for real clients. Built **Maktab**, a multi-tenant school ERP, solo between May and August 2026, and onboarded its first live client within a week of completion. Currently extending that client's website with a multilingual (English, Urdu, Roman Urdu) RAG chatbot. Seeking a full-time full-stack or AI engineering role at a Pakistani company.

EDUCATION

Bachelor of Science in Software Engineering **Expected July 2028**
FAST-NUCES, Lahore, Pakistan | Currently in 5th semester

SYSTEMS SHIPPED & EXPERIENCE

Maktab: Multi-Tenant School ERP / Portal (SaaS) **May 2026 – Present**
Founder & Lead Engineer | *maktab.cloud* | *know.maktab.cloud*

- Designed, built and deployed solo a multi-tenant ERP/portal for K-12 private schools; each school gets its own subdomain on one shared platform (e.g. *bluebells.maktab.cloud*).
- Delivered the teacher portal, assisted admissions, timetable, teacher hiring, sub-admin accounts and role-based access control (RBAC).
- Built with **FastAPI, PostgreSQL, Redis, Celery, Next.js, TypeScript and Tailwind CSS**; deployed with Docker on a Hetzner VPS with Cloudflare R2 storage.

BlueBells School System: Deployment, Website & AI Assistant **2026 – Present**
Client project, Sheikhupura | *bluebells.maktab.cloud* | *bluebellsskp.com*

- First Maktab client, onboarded within a week of completion; **live in production** for about 30 students and 5 teachers.
- Designed and launched the school's public website (bluebellsskp.com).
- Developing an **AI Website Assistant** (launching within weeks): a multilingual RAG chatbot (English, Urdu, Roman Urdu) that answers parents' questions from approved website content with source links. Built with Python, FastAPI and an LLM API.

Furniture Store Website with Admin Backend
Real client project | *furnit-5star.vercel.app*

- Designed the UI and built the storefront and admin backend end to end; deployed on Vercel.

Other Projects

- **QR-Based Attendance System** built with Python.
- **Expense Tracker** with month-wise and category-wise budget tracking.
- **Email Sender** web app using Python, Flask and SMTP.
- **VORTX Gaming Store**: a game vendor website.

TECHNICAL SKILLS

Languages: Python, C++, Java
Backend & Databases: FastAPI, Flask, REST APIs, PostgreSQL, MSSQL, Redis, Celery
Frontend: Next.js, TypeScript, Tailwind CSS
DevOps & Tools: Docker, Hetzner VPS, Cloudflare R2, Git, GitHub, Vercel
AI (currently implementing): LLM APIs and prompt design, RAG, embeddings and semantic search, vector databases, multilingual AI, LangChain, LangGraph
Engineering Practice: Requirements specification, software design and analysis, RBAC, multi-tenant architecture

ACTIVITIES & ACHIEVEMENTS

- **Participant, Softech 2025**: programming competition, FAST-NUCES.
- **Speaker, Career Counselling Session, KIPS Academy, Sheikhupura**: guided students on university admissions, choosing a field by interest and aptitude, and improving the quality of their work.
- **Speaker, Parent Awareness Session, BlueBells School**: led a discussion on how parents and schools can collaborate to create a growth environment for children.